

Antenna Datasheet

Product OC: YE0006AA
Version: 3.4
Date: 2026-05-06
Status: Released

Product Name: 4G Magnetic Mount Whip Monopole External Antenna
Key Features:
Frequency Band: 698–960 MHz, 1710–2690 MHz
Dimensions: Φ 30 mm × 318 mm
Efficiency: Up to 65 % (MP)
RoHS Compliant
IP65

Overview

This Quectel external 4G antenna covers main 4G LTE bands and is compatible with 3G/2G/LPWA bands as well. The external antenna is barely influenced by the internal environment of devices, giving a much better performance in efficiency, radiation and gain whilst providing an optimized solution for a customer product. Quectel also offers flexible installation with custom cable length and connector options.

Contents

Overview **1**
Contents **2**

1 Specification **3**
1.1. Electrical 3
1.2. Mechanical & Environmental 5

2 Drawing **6**

3 Detailed Performance **7**
3.1. S-Parameter Test 7
3.1.1. VSWR 7
3.1.2. Return Loss 8
3.2. Radiation Performance Test 9
3.2.1. Efficiency 9
3.2.2. Average Gain 10
3.2.3. Peak Gain 11
3.2.4. 3D & 2D Radiation Pattern 12
3.2.4.1. Test Condition: On 300 mm × 300 mm Metal Plane 12
3.2.4.2. Test Condition: Free Space 16

4 Packaging **20**

Contact Us **22**
Legal Notices **23**
Revision History **25**

1 Specification

Test Condition: On 300 mm × 300 mm Metal Plane & Free Space

1.1. Electrical

Electrical	
Frequency Range	698–960 MHz, 1710–2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

Electrical – Detail

Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR	MP	-	3.0	3.0	-	2.9	2.1	2.5	2.4	-	-	-
	FS	-	3.2	2.7	-	2.6	2.0	2.6	2.5	-	-	-
Max. Return Loss (dB)	MP	-	-6.0	-6.1	-	-6.3	-8.8	-7.4	-7.7	-	-	-
	FS	-	-5.7	-6.8	-	-6.9	-9.3	-6.9	-7.4	-	-	-
AVG Eff. (%)	MP	-	46.2	53.5	-	33.6	28.3	27.1	35.3	-	-	-
	FS	-	24.4	42.9	-	33.8	26.7	23.7	33.7	-	-	-
AVG AVG Gain (dB)	MP	-	-3.4	-2.8	-	-4.8	-5.5	-5.7	-4.6	-	-	-
	FS	-	-6.1	-3.7	-	-4.8	-5.7	-6.2	-4.8	-	-	-
Max. Peak Gain (dBi)	MP	-	0.1	0.7	-	4.5	0.8	1.2	3.0	-	-	-
	FS	-	-2.1	2.6	-	4.9	-0.9	-0.8	2.6	-	-	-

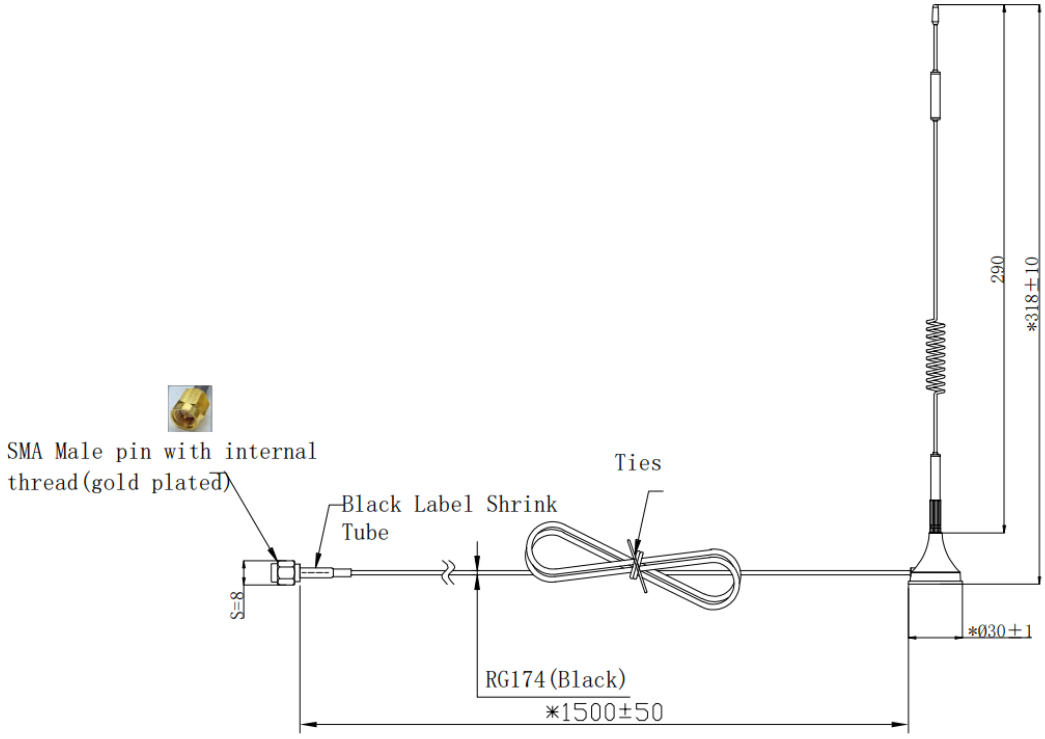
VSWR	MP	≤ 3
	FS	≤ 3.2
Return Loss	MP	≤ -6 dB
	FS	≤ -5.7 dB
Peak Gain	MP	≤ 4.5 dBi
	FS	≤ 4.9 dBi

- FS: Free Space
- MP: On 300 mm × 300 mm Metal Plane

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	Φ 30 mm × 318 mm
Material & Color	Carbon Steel & Black
Cable Type & Color & Length	RG 174 & Black & 1500 mm
Connector Type	SMA Male
Mounting Type	Magnet
Weight	Typ. 41.8 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
Ingress Protection (IP) Rating	IP65
RoHS Compliant	Yes

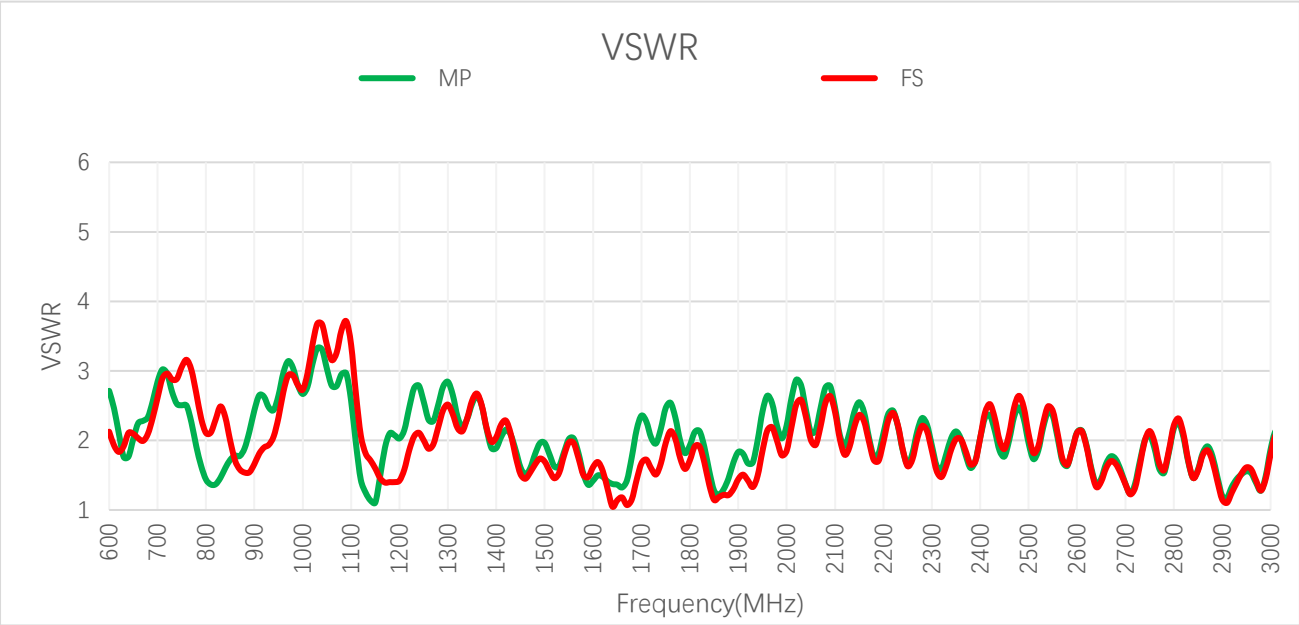
2 Drawing


SMA Male pin with internal thread(gold plated)
Black Label Shrink Tube
Ties
290
*318±10
S=8
*Ø30±1
RG174(Black)
*1500±50

3 Detailed Performance

3.1. S-Parameter Test

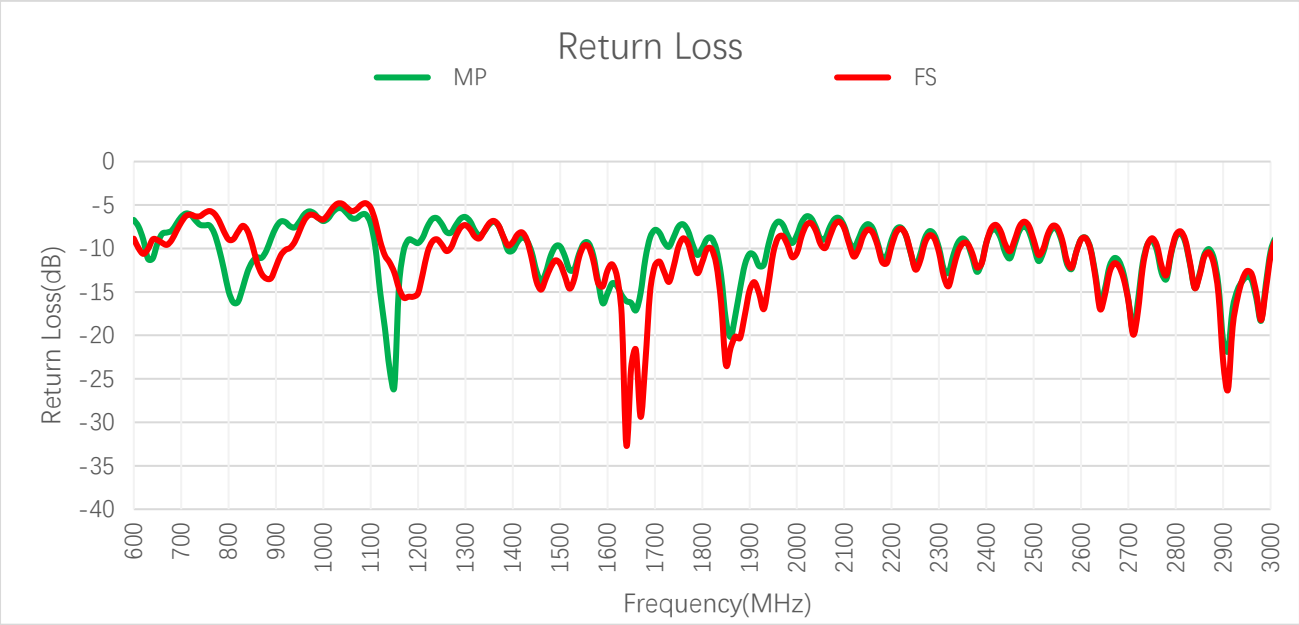
3.1.1. VSWR



VSWR

Frequency (MHz)		600	630	710	830	900	960	1440	1710	1740	1880
VSWR	MP	-	-	3.0	1.5	2.4	3.0	-	2.3	2.2	1.5
	FS	-	-	2.9	2.5	1.7	2.7	-	1.7	1.7	1.2
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
VSWR	MP	2.4	2.4	2.1	1.8	2.1	1.6	-	-	-	-
	FS	1.8	2.2	2.0	1.9	2.1	1.5	-	-	-	-

3.1.2. Return Loss

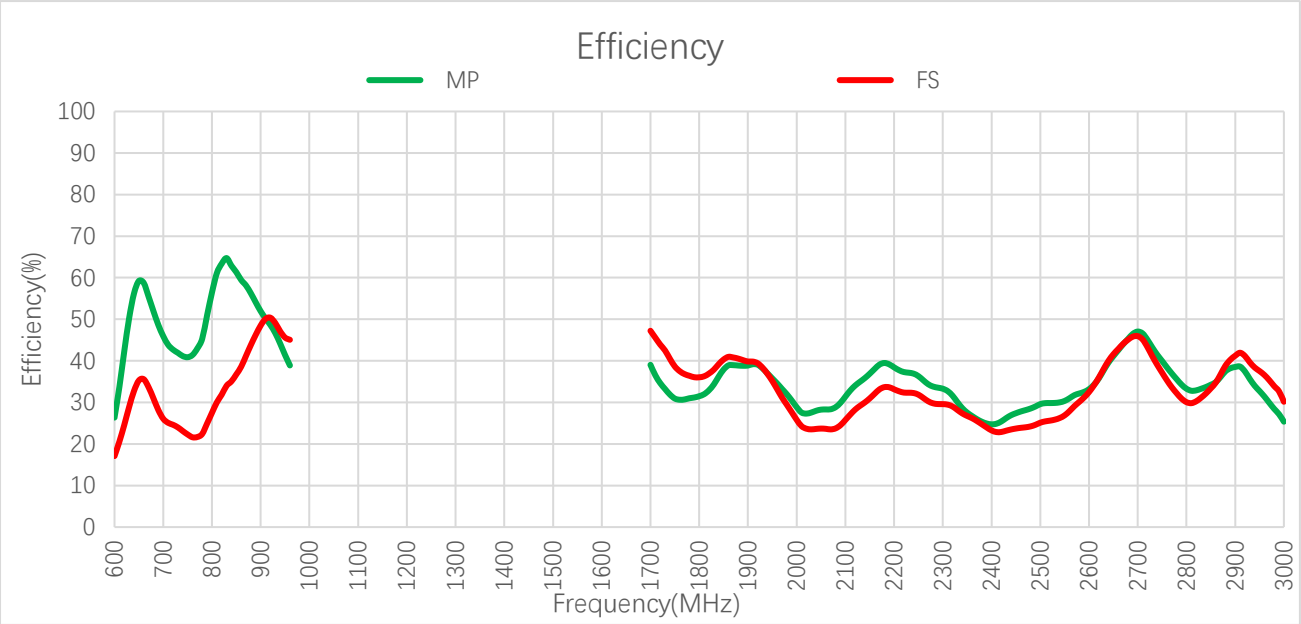


Return Loss (dB)

Frequency (MHz)		600	630	710	830	900	960	1440	1710	1740	1880
Return Loss (dB)	MP	-	-	-6.0	-14.5	-7.6	-6.1	-	-8.2	-8.7	-14.6
	FS	-	-	-6.3	-7.4	-12.1	-6.8	-	-11.5	-12.0	-20.3
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Return Loss (dB)	MP	-7.7	-7.7	-8.8	-11.1	-8.9	-13.0	-	-	-	-
	FS	-10.5	-8.5	-9.4	-10.3	-9.0	-13.5	-	-	-	-

3.2. Radiation Performance Test

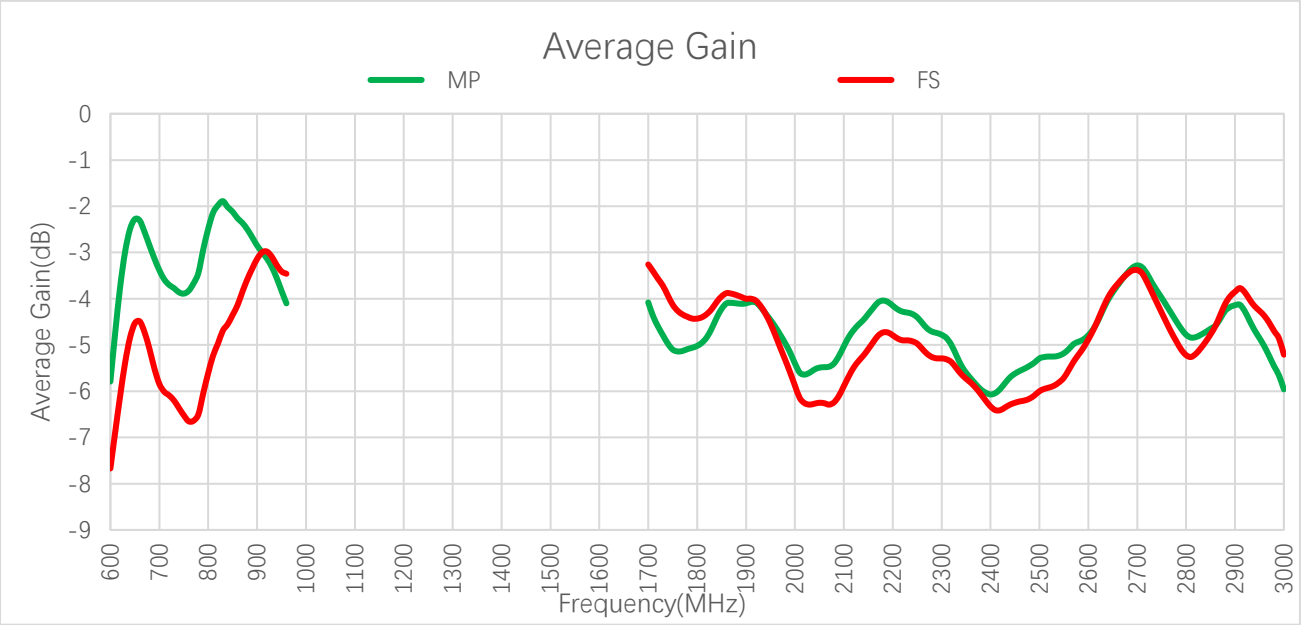
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)		600	630	710	830	900	960	1440	1710	1740	1880
Efficiency (%)	MP	-	-	43.9	64.7	51.9	38.9	-	36.5	31.9	38.9
	FS	-	-	25.1	34.0	48.5	45.1	-	45.6	40.5	40.5
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Efficiency (%)	MP	35.8	35.9	27.5	27.4	33.2	46.5	-	-	-	-
	FS	35.2	30.0	26.7	23.7	32.5	45.8	-	-	-	-

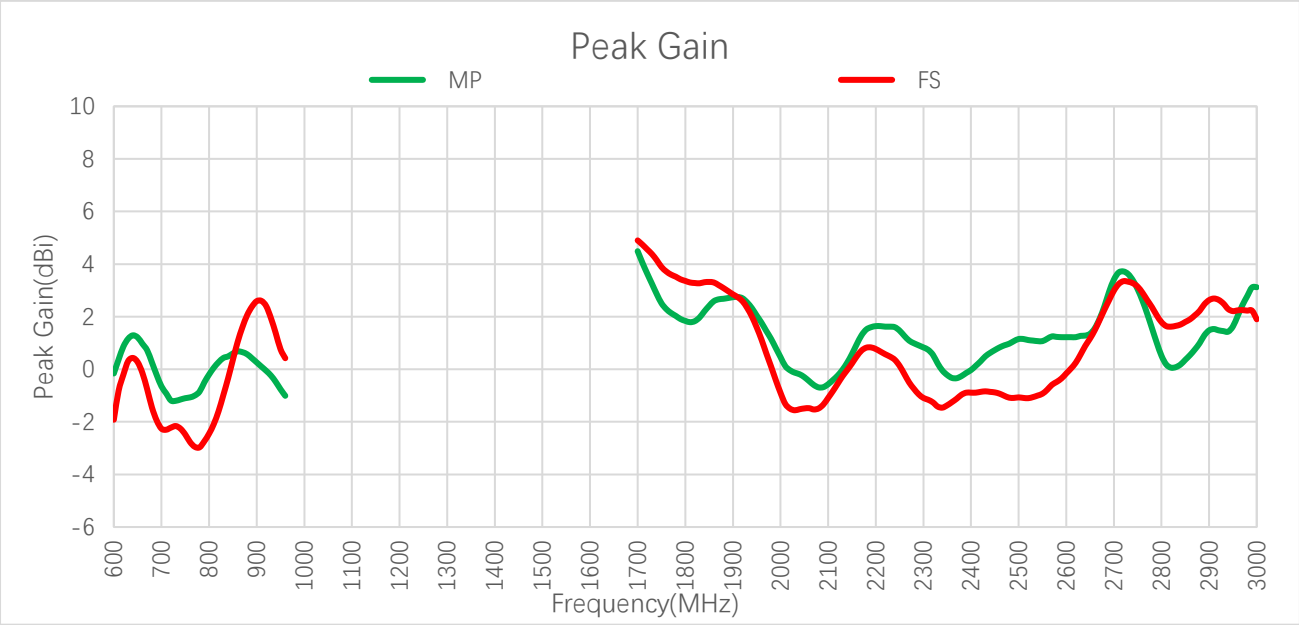
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)		600	630	710	830	900	960	1440	1710	1740	1880
Average Gain (dB)	MP	-	-	-3.6	-1.9	-2.9	-4.1	-	-4.4	-5.0	-4.1
	FS	-	-	-6.0	-4.7	-3.1	-3.5	-	-3.4	-3.9	-3.9
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Average Gain (dB)	MP	-4.5	-4.5	-5.6	-5.6	-4.8	-3.3	-	-	-	-
	FS	-4.5	-5.2	-5.7	-6.3	-4.9	-3.4	-	-	-	-

3.2.3. Peak Gain



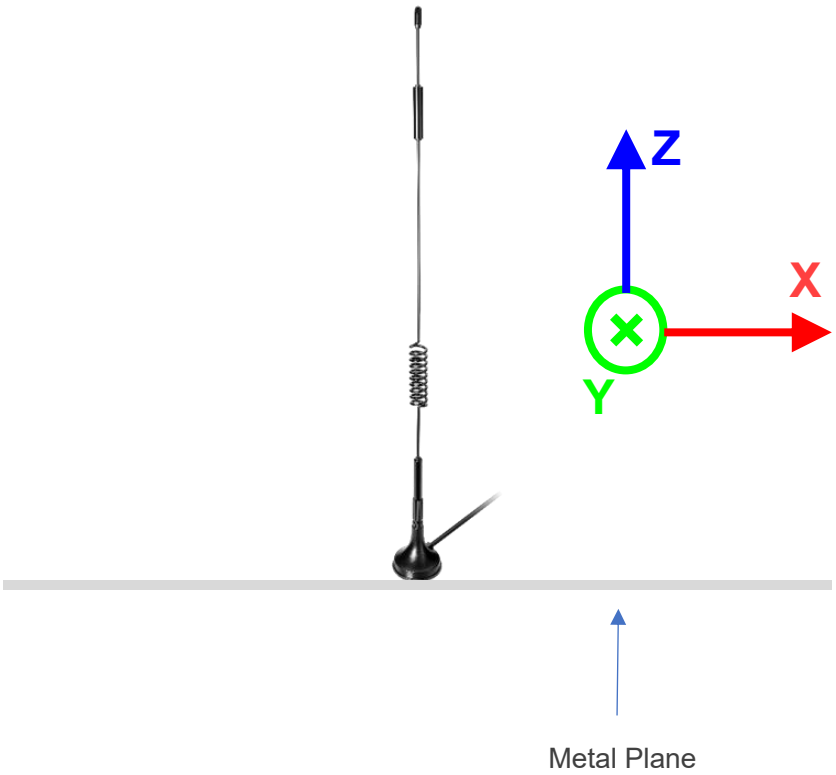
Peak Gain (dBi)

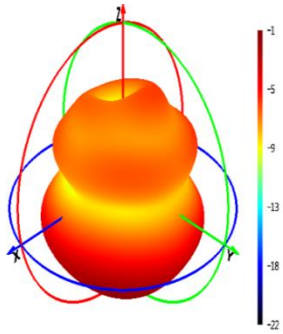
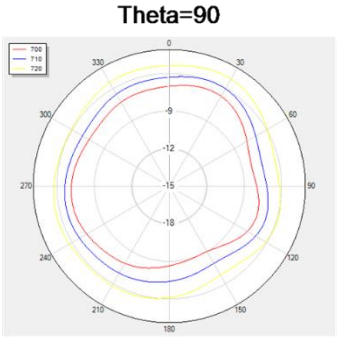
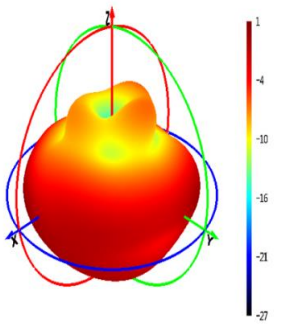
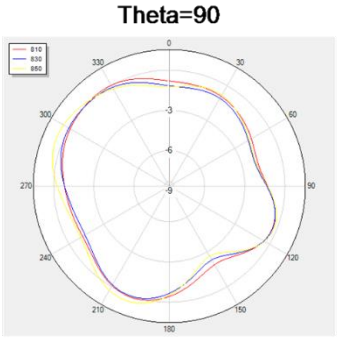
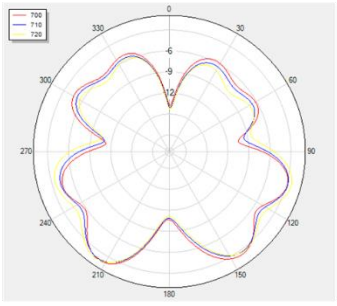
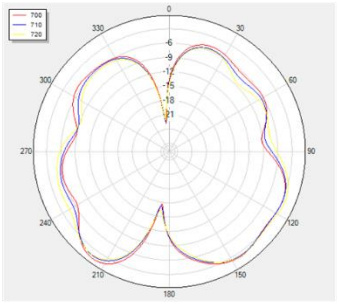
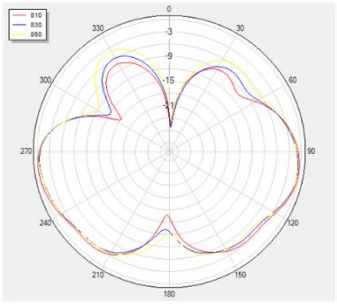
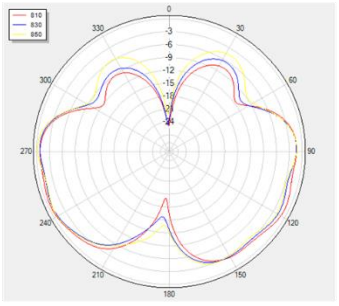
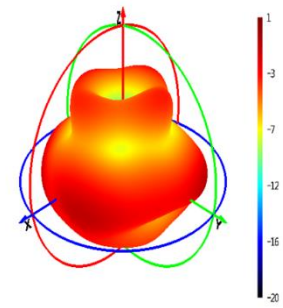
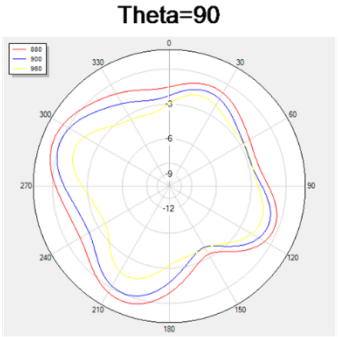
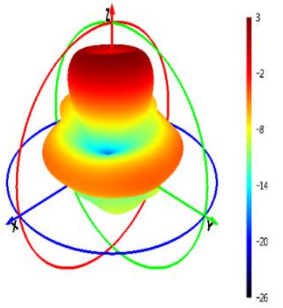
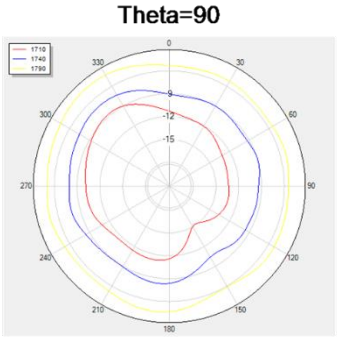
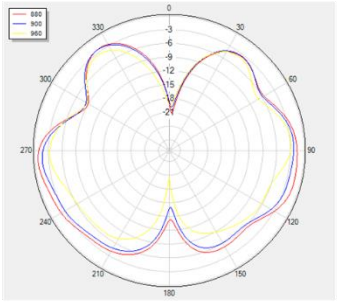
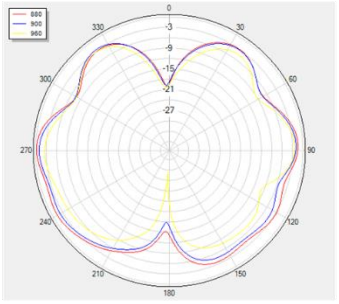
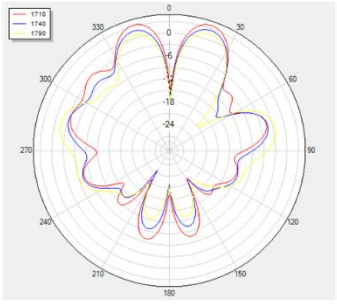
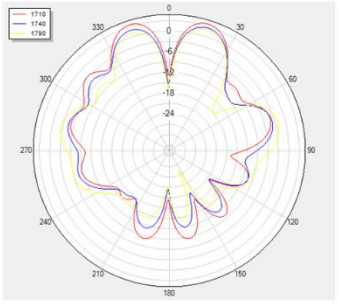
Frequency (MHz)		600	630	710	830	900	960	1440	1710	1740	1880
Peak Gain (dBi)	MP	-	-	-0.9	0.4	0.3	-1.0	-	4.1	2.9	2.7
	FS	-	-	-2.3	-1.0	2.6	0.4	-	4.7	4.2	3.1
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Peak Gain (dBi)	MP	2.0	0.3	-0.2	0.7	1.2	3.0	-	-	-	-
	FS	1.6	0.0	-1.4	-0.9	-0.2	2.6	-	-	-	-

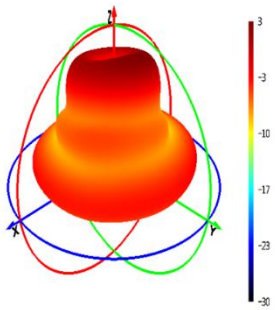
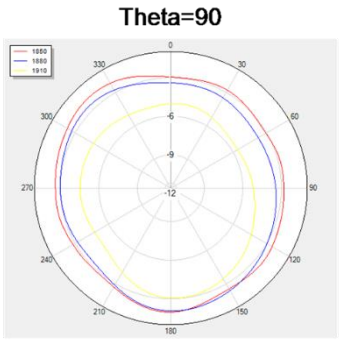
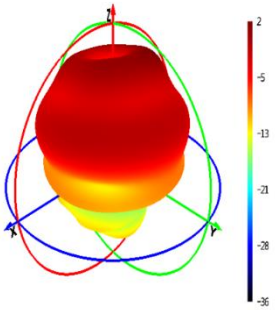
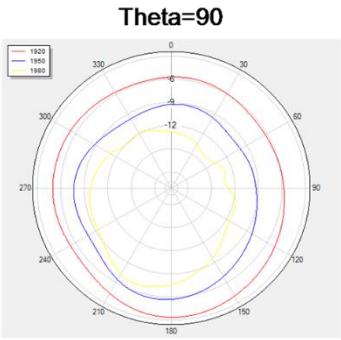
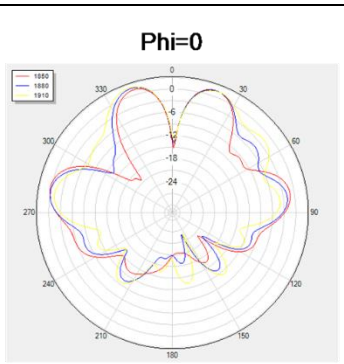
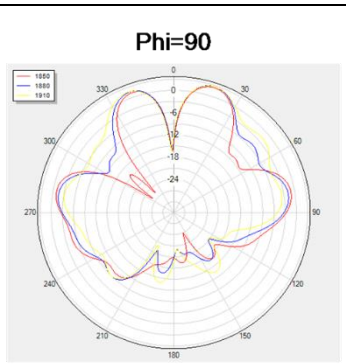
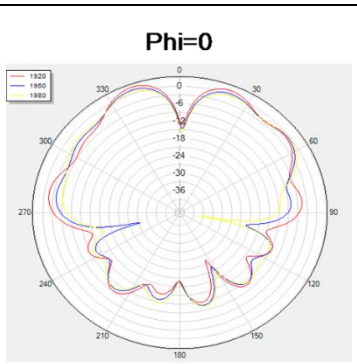
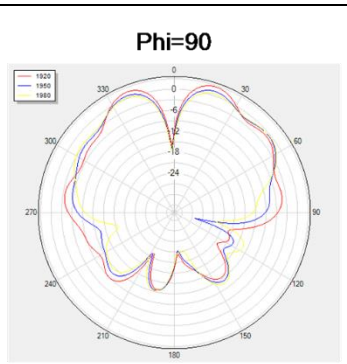
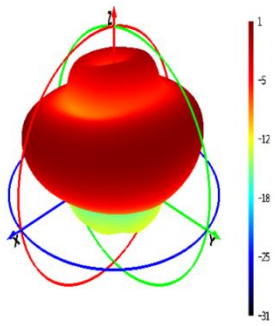
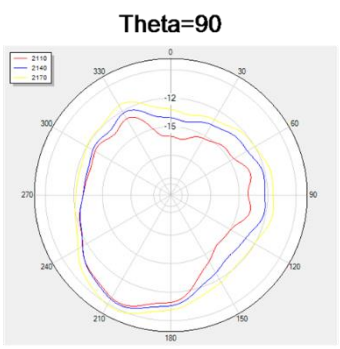
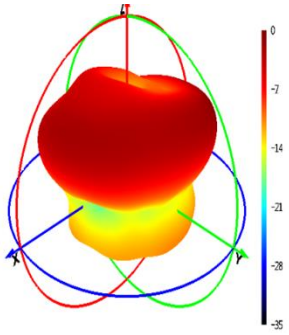
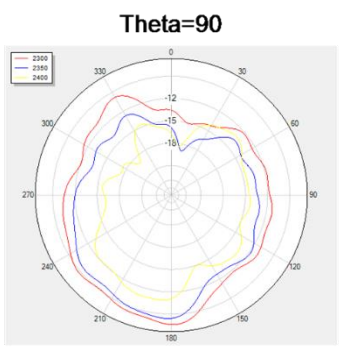
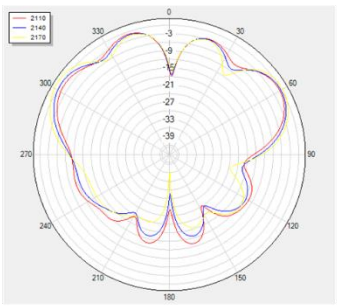
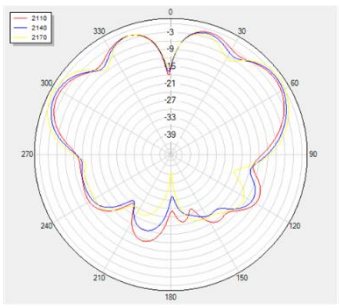
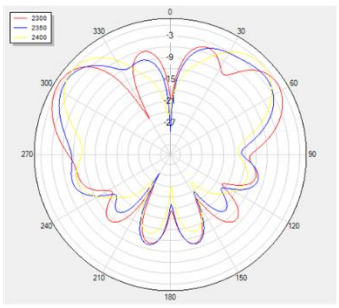
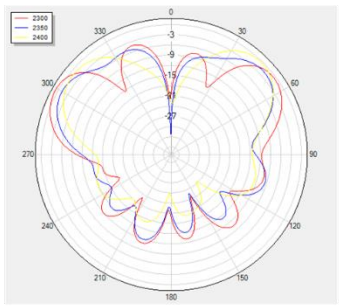
3.2.4. 3D & 2D Radiation Pattern

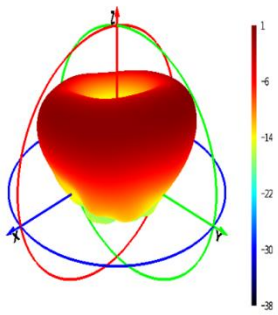
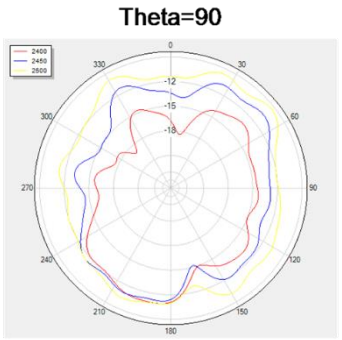
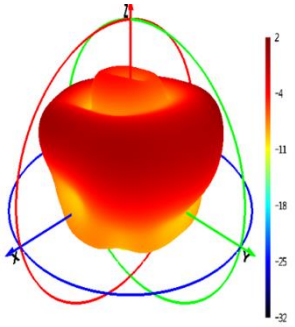
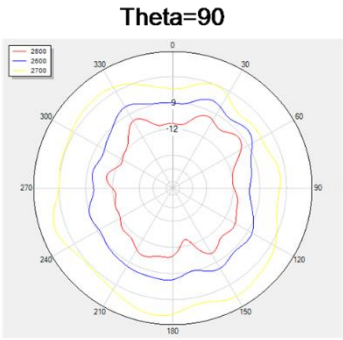
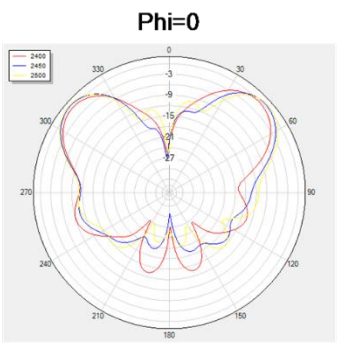
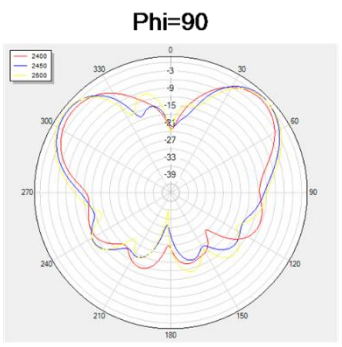
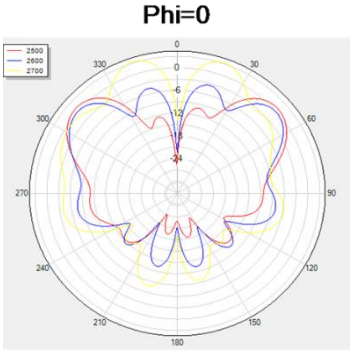
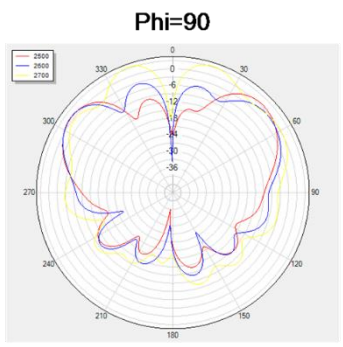
3.2.4.1. Test Condition: On 300 mm × 300 mm Metal Plane

- Test Chamber: GL-S-1



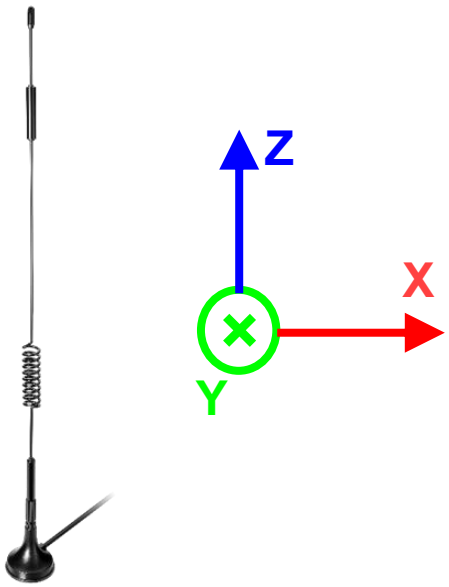
710 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

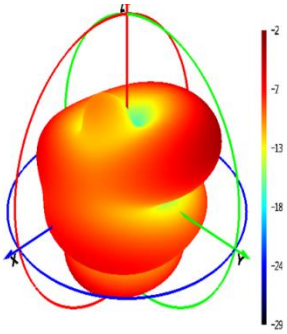
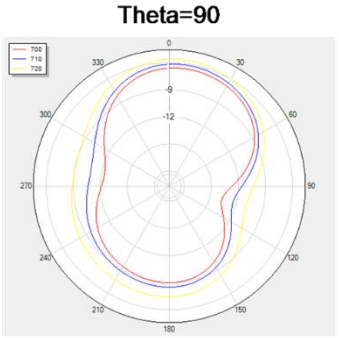
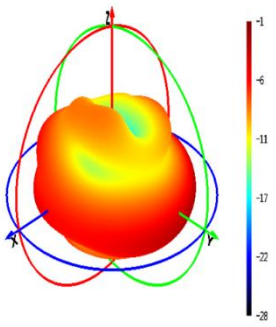
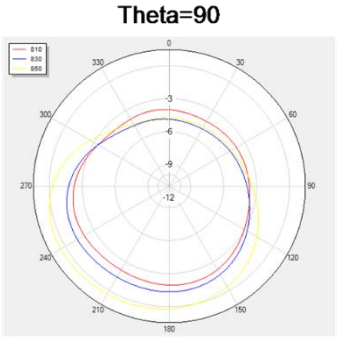
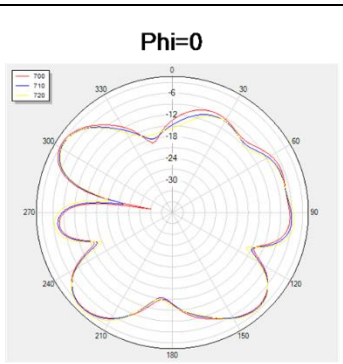
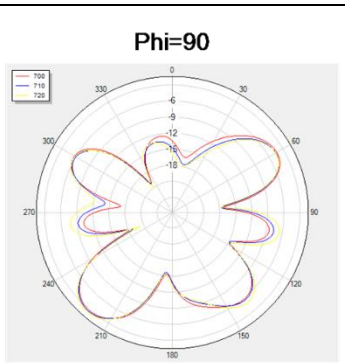
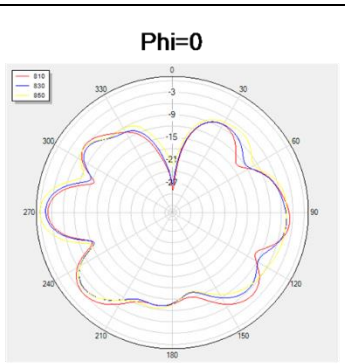
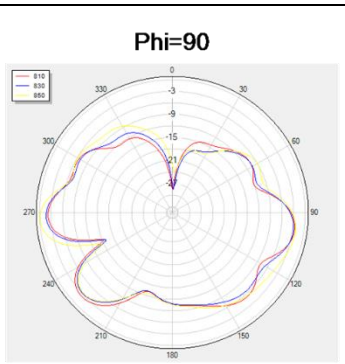
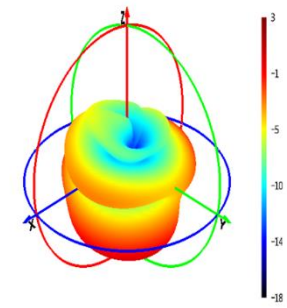
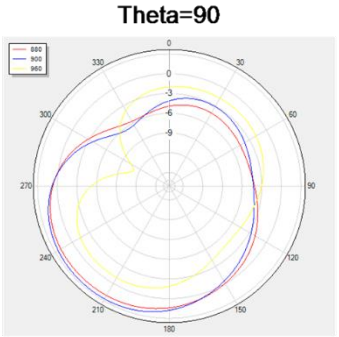
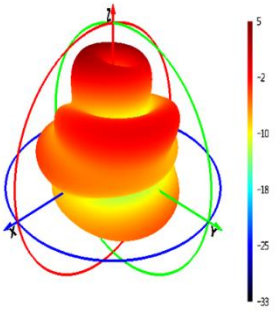
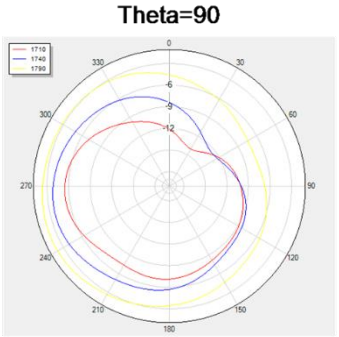
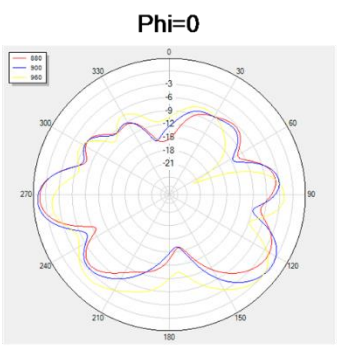
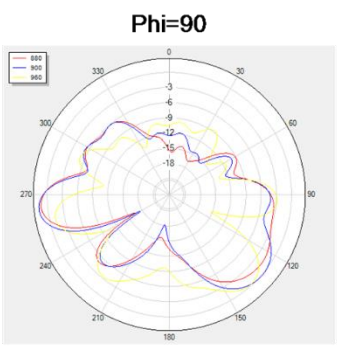
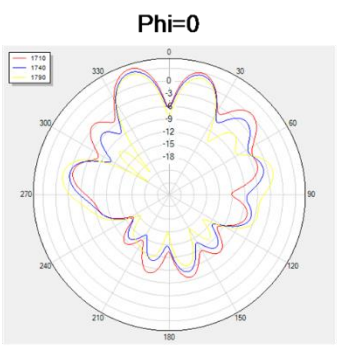
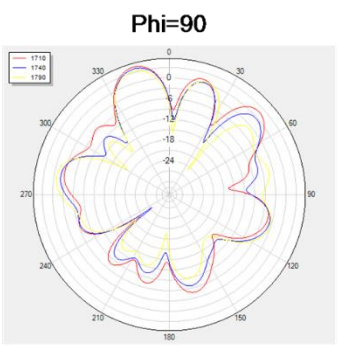
1880 MHz		1950 MHz	
			
			
2140 MHz		2350 MHz	
			
			

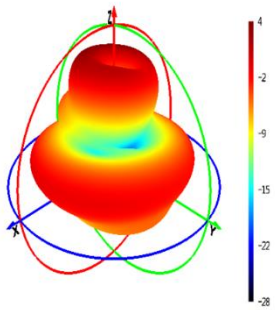
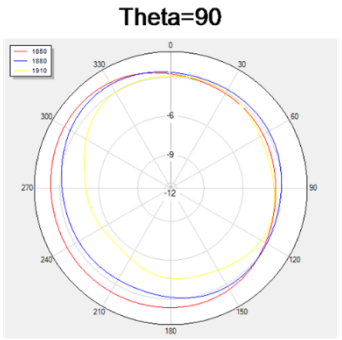
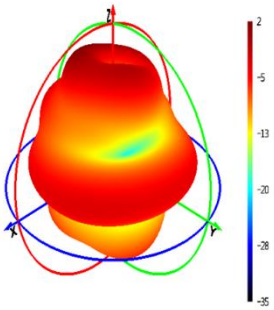
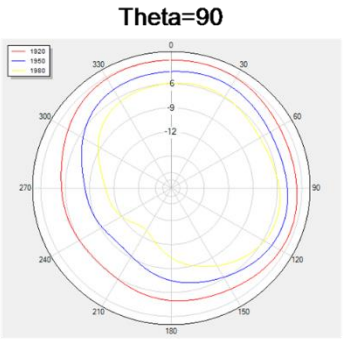
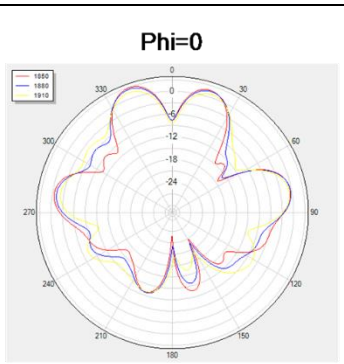
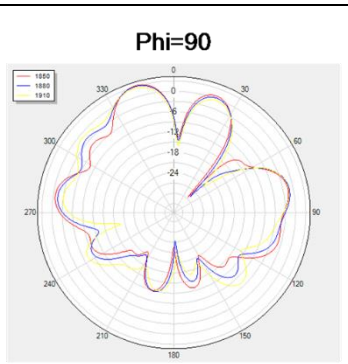
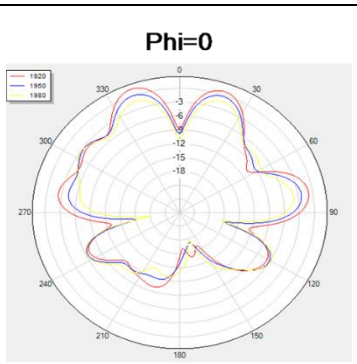
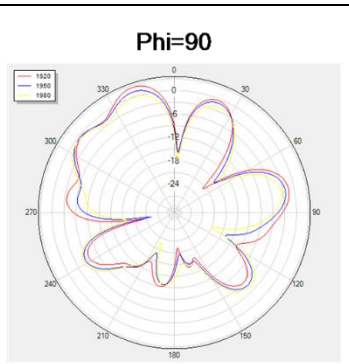
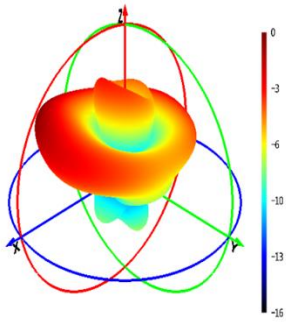
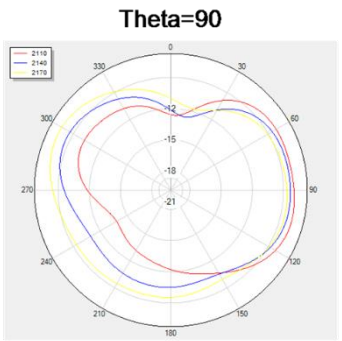
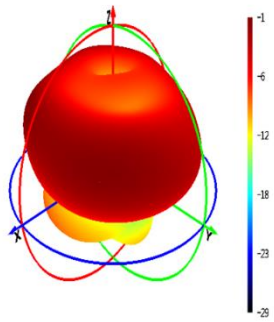
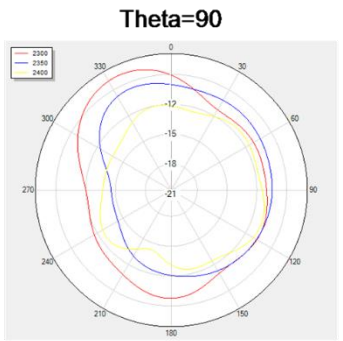
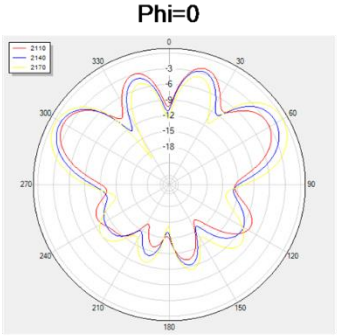
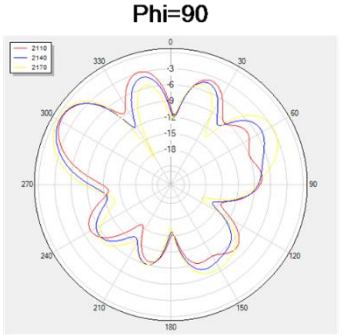
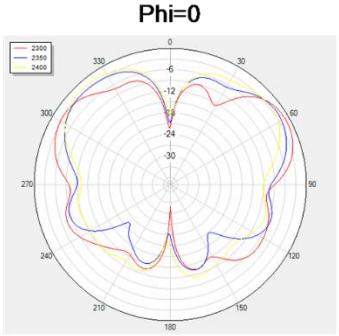
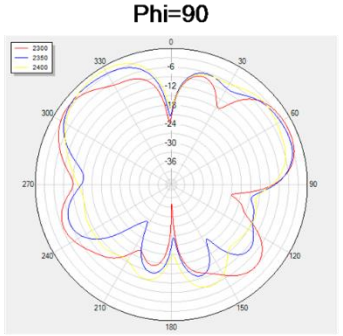
2450 MHz		2600 MHz	
			
			

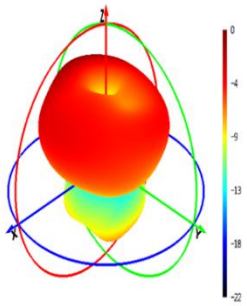
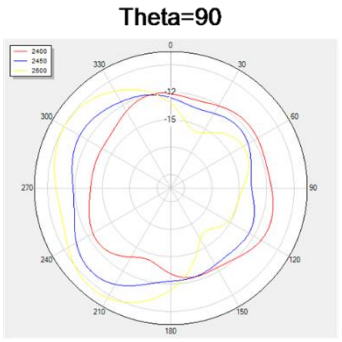
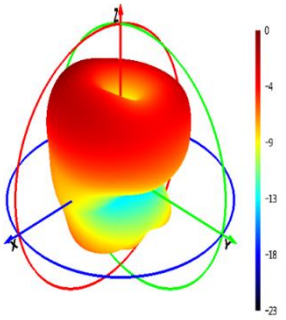
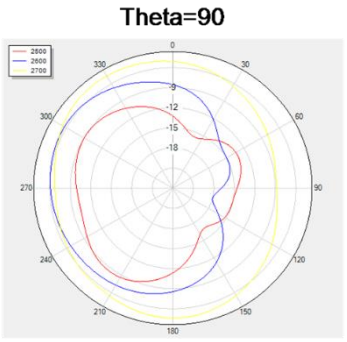
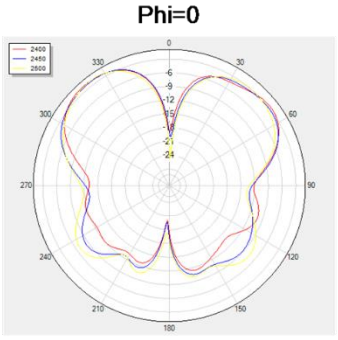
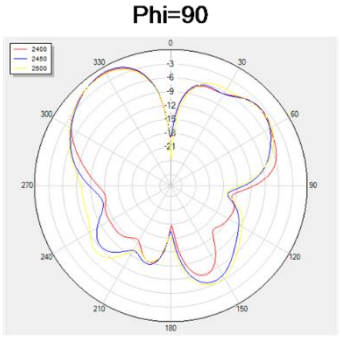
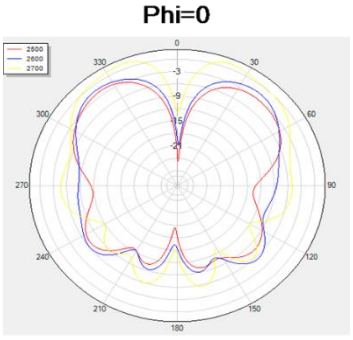
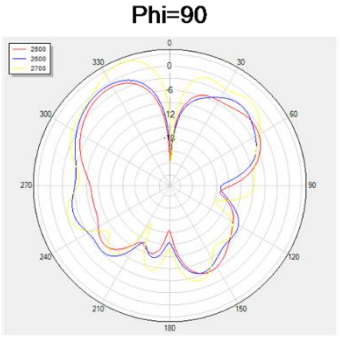
3.2.4.2. Test Condition: Free Space

- Test Chamber: GL-S-1

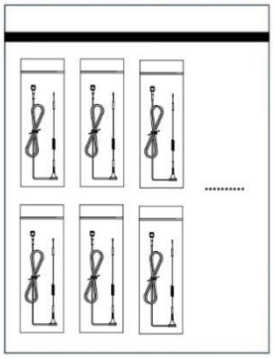
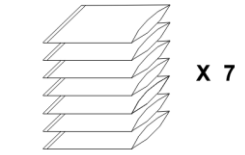
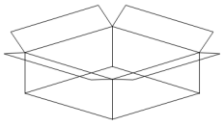
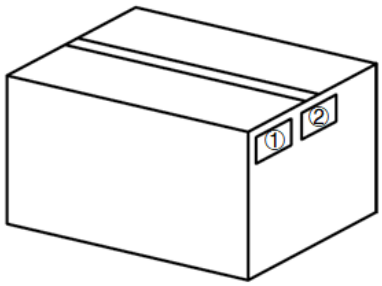


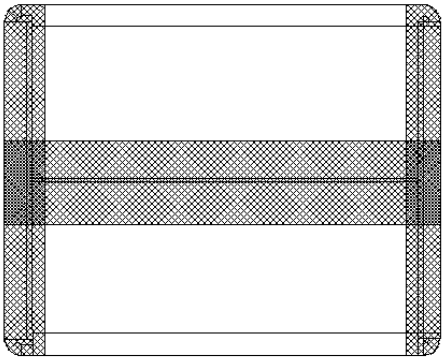
710 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

1880 MHz		1950 MHz	
			
			
2140 MHz		2350 MHz	
			
			

2450 MHz		2600 MHz	
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		1 antenna products in a small PE bag. (1 antenna / Small PE Bag)
2	 X 30	30 antenna products in a big PE bag. (30 antennas / Big PE Bag)
3	 X 7 	(7 Big PE Bags / Carton Box) (210 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 370 × 370 × 295 mm
4		**Position for Attaching Labels** ① Carton Label ② Quality Label

5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
https://www.quectel.com/tech-support/.
Or email us at: support@quectel.com.

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Revision History

Version	Date	Author	Note
-	2021-04-14	Kenny Yin	Creation of the document
1.0	2020-06-16	Kenny Yin	First official release
1.1	2020-12-11	Kenny Yin	Updated the antenna image (Chapter 2).
1.2	2021-01-12	Kenny Yin	Updated the efficiency data (Chapter 4).
1.3	2021-01-27	Kenny Yin	Added IP rating description and installation method.
2.0	2021-04-30	Aria Chu	Updated all test data in the datasheet.
2.1	2021-07-25	Aria Chu	1. Updated working temperature (Chapter 3). 2. Added detailed passive electrical specifications (Chapter 3).
2.2	2021-12-01	Aria Chu	Updated the product description (Chapter 1).
3.0	2023-02-24	Damon Zhang/ Lucky Feng/ David Liu/ Aria Chu	Updated all data and datasheet template.
3.1	2023-08-15	Lucky Feng	Updated the drawing (Chapter 2).
3.2	2025-03-11	Rainey Liao	Updated the starting frequency to 698 MHz (Homepage and Chapter 1.1).
3.3	2025-04-24	Rainey Liao	Updated the antenna image (Cover page).
3.4	2026-05-06	Kessie Geng	Updated the packaging (Chapter 4).



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